

Q1 - 25 July - Shift 1

Let the equation of two diameters of a circle $x^2 + y^2 - 2x + 2fy + 1 = 0$ be $2px - y = 1$ and $2x + py = 4p$. Then the slope $m \in (0, \infty)$ of the tangent to the hyperbola $3x^2 - y^2 = 3$ passing through the centre of the circle is equal to _____.

Space for your notes:

Q2 - 25 July - Shift 2

Let the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{7} = 1$ and the

Space for your notes:

hyperbola $\frac{x^2}{144} - \frac{y^2}{\alpha} = \frac{1}{25}$ coincide. Then the length of the latus rectum of the hyperbola is:-

(A) $\frac{32}{9}$

(B) $\frac{18}{5}$

(C) $\frac{27}{4}$

(D) $\frac{27}{10}$

Q3 - 26 July - Shift 1

Let the tangent drawn to the parabola $y^2 = 24x$ at the point (α, β) is perpendicular to the line $2x + 2y = 5$. Then the normal to the hyperbola

Space for your notes:

$\frac{x^2}{\alpha^2} - \frac{y^2}{\beta^2} = 1$ at the point $(\alpha + 4, \beta + 4)$ does NOT pass through the point :

(A) (25, 10)

(B) (20, 12)

(C) (30, 8)

(D) (15, 13)

Q4 - 26 July - Shift 2

If the line $x-1=0$, is a directrix of the hyperbola $kx^2 - y^2 = 6$, then the hyperbola passes through the point

(A) $(-2\sqrt{5}, 6)$ (B) $(-\sqrt{5}, 3)$

(C) $(\sqrt{5}, -2)$ (D) $(2\sqrt{5}, 3\sqrt{6})$

Space for your notes:

Q5 - 27 July - Shift 1

An ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through the

vertices of the hyperbola $H: \frac{x^2}{49} - \frac{y^2}{64} = -1$. Let

the major and minor axes of the ellipse E coincide with the transverse and conjugate axes of the hyperbola H. Let the product of the eccentricities

of E and H be $\frac{1}{2}$. If l is the length of the latus

rectum of the ellipse E, then the value of $113l$ is equal to _____.

Space for your notes:

Q6 - 27 July - Shift 2

A common tangent T to the curves

$C_1: \frac{x^2}{4} + \frac{y^2}{9} = 1$ and $C_2: \frac{x^2}{42} - \frac{y^2}{143} = 1$ does not

pass through the fourth quadrant. If T touches C_1 at (x_1, y_1) and C_2 at (x_2, y_2) , then $|2x_1 + x_2|$ is equal to

_____.

Space for your notes:

Q7 - 28 July - Shift 1

For the hyperbola $H : x^2 - y^2 = 1$ and the ellipse

$$E : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b > 0, \text{ let the}$$

(1) eccentricity of E be reciprocal of the eccentricity of H , and

(2) the line $y = \sqrt{\frac{5}{2}}x + K$ be a common tangent of E and H .

Then $4(a^2 + b^2)$ is equal to _____.

Space for your notes:

Q8 - 28 July - Shift 2

Let the hyperbola $H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ pass through

the point $(2\sqrt{2}, -2\sqrt{2})$. A parabola is drawn

whose focus is same as the focus of H with positive abscissa and the directrix of the parabola passes through the other focus of H . If the length of the latus rectum of the parabola is e times the length of the latus rectum of H , where e is the eccentricity of H , then which of the following points lies on the parabola?

Space for your notes:

Q9 - 29 July - Shift 1

Let the focal chord of the parabola $P : y^2 = 4x$ along the line $L : y = mx + c$, $m > 0$ meet the parabola at the points M and N. Let the line L be a tangent to the hyperbola $H : x^2 - y^2 = 4$. If O is the vertex of P and F is the focus of H on the positive x-axis, then the area of the quadrilateral OMFN is :

- (A) $2\sqrt{6}$ (B) $2\sqrt{14}$
(C) $4\sqrt{6}$ (D) $4\sqrt{14}$

Space for your notes:

Answer Key**Q1 (2)****Q2 (D)****Q3 (D)****Q4 (C)****Q5 (1552)****Q6 (20)****Q7 (3)****Q8 (B)****Q9 (B)**

#MathBoleTohMathonGo

Q1 (2)

$$2p + f - 1 = 0 \quad \dots(1)$$

$$2 - pf - 4p = 0 \quad \dots(2)$$

$$2 = p(f + 4)$$

$$p = \frac{2}{f + 4}$$

$$2p = 1 - f$$

$$\frac{4}{f + 4} = 1 - f$$

$$f^2 + 3f = 0$$

$$f = 0 \text{ or } -3$$

$$\text{Hyperbola } 3x^2 - y^2 = 3, \quad x^2 - \frac{y^2}{3} = 1$$

$$y = mx \pm \sqrt{m^2 - 3}$$

It passes (1, 0)

$$0 = m \pm \sqrt{m^2 - 3}$$

m tends ∞

It passes (1, 3)

$$3 = m \pm \sqrt{m^2 - 3}$$

$$(3 - m)^2 = m^2 - 3$$

$$m = 2$$

Q2 (D)

(D)

Q3 (D)Tangent at (α, β) has slope 1

$$\beta^2 = 24\alpha$$

Equation of tangent $y\beta = 12(x + \alpha)$, $\frac{12}{\beta} = 1$

$$\Rightarrow \alpha = 6, \beta = 12$$

$$\therefore (\alpha + 4, \beta + 4) = (10, 16)$$

Normal at $(10, 16)$ to $\frac{x^2}{36} - \frac{y^2}{144} = 1$ is

$$2x + 5y = 100$$

Q4 (C)

$$\frac{x^2}{6/k} - \frac{y^2}{6} = 1 \quad \dots\dots(1)$$

$$e^2 = 1 + \frac{6}{6/k}$$

$$e = \sqrt{1+k}$$

$$a = \sqrt{\frac{6}{k}}$$

$$\text{Eq. of directrix } x = \frac{a}{e} \Rightarrow x = \sqrt{\frac{6}{k(k+1)}}$$

$$\frac{6}{k(k+1)} = 1$$

$$k = 2$$

$$\text{From eq. (1), we get } 2x^2 - y^2 = 6$$

Check options

Q5 (1552)

$$\text{Hyp: } \frac{y^2}{64} - \frac{x^2}{49} = 1$$

An ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ passes through the

vertices of the hyperbola $H: \frac{x^2}{49} - \frac{y^2}{64} = -1$.

$$\text{So } b^2 = 64$$

$$e_H = \sqrt{1 + \frac{a^2}{b^2}} = \sqrt{1 + \frac{49}{64}}$$

$$\text{Ellipse } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$e_E = \sqrt{1 - \frac{a^2}{b^2}} = \sqrt{1 - \frac{a^2}{64}}$$

$$b = 8, \sqrt{\frac{1-a^2}{64}} \times \frac{\sqrt{113}}{8} = \frac{1}{2} \Rightarrow \sqrt{64-a^2} \times \sqrt{113} = 32$$

$$(64-a^2) = \frac{32^2}{113}$$

$$\Rightarrow a^2 = 64 - \frac{32^2}{113}$$

$$l = \frac{2a^2}{b} = \frac{2}{8} \left(64 - \frac{32^2}{113} \right) = \frac{1552}{113}$$

$$113l = 1552$$

Q6 (20)

Let common tangents are

$$T_1 : y = mx \pm \sqrt{4m^2 + 9}$$

$$\& T_2 : y = mx \pm \sqrt{42m^2 - 13}$$

$$\text{So, } 4m^2 + 9 = 42m^2 - 143$$

$$\Rightarrow 38m^2 = 152$$

$$\Rightarrow m = \pm 2$$

$$\& c = \pm 5$$

For given tangent not pass through 4th quadrant

$$T : y = 2x + 5$$

$$\text{Now, comparing with } \frac{xx_1}{4} + \frac{yy_1}{9} = 1$$

$$\text{We get, } \frac{x_1}{8} = -\frac{1}{5} \Rightarrow x_1 = -\frac{8}{5}$$

$$\frac{xx_2}{42} - \frac{yy_2}{143} = 1$$

$$2x - y = -5 \text{ we have}$$

$$x_2 = -\frac{84}{5}$$

$$\text{So, } |2x_1 + x_2| = \left| \frac{-100}{5} \right| = 20$$

#MathBoleTohMathonGo

To practice more chapter-wise JEE Main PYQs, [click here to download the MARKS app](#) from Playstore

$$e_E = \sqrt{1 - \frac{b^2}{a^2}}, e_H = \sqrt{2}$$

$$\text{If } \Rightarrow e_E = \frac{1}{e_H}$$

$$\Rightarrow \frac{a^2 - b^2}{a^2} = \frac{1}{2}$$

$$2a^2 - 2b^2 = a^2$$

$$a^2 = 2b^2$$

and $y = \sqrt{\frac{5}{2}}x + k$ is tangent to ellipse then

$$K^2 = a^2 \times \frac{5}{2} + b^2 = \frac{3}{2}$$

$$6b^2 = \frac{3}{2} \Rightarrow b^2 = \frac{1}{4} \text{ and } a^2 = \frac{1}{2}$$

$$\therefore 4.(a^2 + b^2) = 3$$

Q8 (B)

$$H : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Foci : S (ae, 0), S' (-ae, 0)

Foot of directrix of parabola is (-ae, 0)

Focus of parabola is (ae, 0)

Now, semi latus rectum of parabola = |SS'| = 2ae

$$\text{Given, } 4ae = e \left(\frac{2b^2}{a} \right)$$

$$\Rightarrow b^2 = 2a^2 \quad \dots (1)$$

Given, $(2\sqrt{2}, -2\sqrt{2})$ lies on H

$$\Rightarrow \frac{1}{a^2} - \frac{1}{b^2} = \frac{1}{8} \quad \dots (2)$$

From (1) and (2)

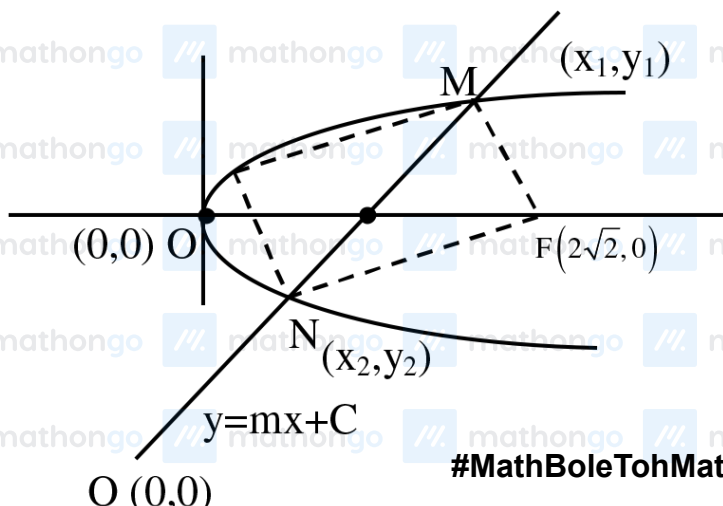
$$a^2 = 4, b^2 = 8$$

$$\therefore b^2 = a^2(e^2 - 1)$$

$$\therefore e = \sqrt{3}$$

$\Rightarrow$ Equation of parabola is $y^2 = 8\sqrt{3}x$

Q9 (B)



#MathBoleTohMathonGo

Hyperbola

Hints and Solutions

Focus (ae, 0)

$$F(2\sqrt{2}, 0)$$

Line L : y = mx + c pass (1,0)

$$0 = m + C \dots\dots(1)$$

Line L is tangent to Hyperbola. $\frac{x^2}{4} - \frac{y^2}{4} = 1$

$$C = \pm \sqrt{a^2 m^2 - \ell^2}$$

$$C = \pm \sqrt{4m^2 - 4}$$

From (1)

$$-m = \pm \sqrt{4m^2 - 4}$$

Squaring

$$m^2 = 4m^2 - 4$$

$$4 = 3m^2$$

$$\frac{2}{\sqrt{3}} = m \text{ (as } m > 0)$$

$$C = -m$$

$$C = \frac{-2}{\sqrt{3}}$$

$$y = \frac{2x}{\sqrt{3}} - \frac{2}{\sqrt{3}}$$

$$y^2 = 4x$$

$$\Rightarrow \left(\frac{2x - 2}{\sqrt{3}} \right)^2 = 4x$$

$$\Rightarrow x^2 + 1 - 2x = 3x$$

$$\Rightarrow x^2 - 5x + 1 = 0$$

$$y^2 = 4 \left(\frac{\sqrt{3}y + 2}{2} \right)$$

#MathBoleTohMathonGo

$$\Rightarrow y^2 - 2\sqrt{3}y - 4 = 0$$

Area

$$\left| \begin{array}{ccccc} 1 & 0 & x_1 & 2\sqrt{2} & x_2 & 0 \\ 2 & 0 & y_1 & 0 & y_2 & 0 \end{array} \right|$$

$$= \left| \frac{1}{2} [-2\sqrt{2}y_1 + 2\sqrt{2}y_2] \right|$$

$$= \sqrt{2} |y_2 - y_1| = \frac{(\sqrt{2})\sqrt{12+16}}{111}$$

$$= \sqrt{56}$$

$$= 2\sqrt{14}$$